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Module 9

CompTIA Security+ Exam Objectives

Domain	**Lab**
General Security Concepts	9.5
Threats, Vulnerabilities, and Mitigation	9.1, 9.2
Security Architecture	9.4
Security Operations	9.1, 9.2, 9.3

Website links change frequently. If a link in the following activities no longer works, use a search engine such as Google or Bing to search for the information.

Lab 9.5 Zero Trust—Trust No One, Verify Everything

Estimated time: 30 minutes

Objectives

The Zero Trust model is based on the premise of "Never trust, always verify." It is also known as "perimeterless" security, Zero Trust Architecture (ZTA), Zero Trust Network Architecture, and Zero Trust Network Access (ZTNA). In the new era of information technology and the increasing use of environments and infrastructures that go beyond on-premises networks, cybersecurity has become a much tougher component to implement. Multicloud-enabled solutions, the Internet of Things, data analytics, machine learning, and artificial intelligence have added a plethora of new endpoints to protect, including the attack surface or network perimeter (network boundaries). A network's new perimeter or boundary has expanded and become blurred with the increased use of the cloud. Implementing a strong Zero Trust model can help your infrastructure be better protected and withstand an attack.

Following are Zero Trust's guiding principles:

- Verify explicitly
- Least privileged access
- Assume breach

Its foundational pillars include the following:

- Identities may be users, services, or devices.
- Devices create a large attack surface as data flows.
- Applications are the way that data is consumed.
- Data should be classified, labeled, and encrypted based on its attributes.
- Infrastructure, whether on-premises or cloud based, represents a threat vector.
- Networks should be segmented.

After completing this lab, you will be able to do the following:

- Define a Zero Trust model.
- Identify key components of a Zero Trust model.

Certification Objectives

Objectives for the CompTIA Security+ exam:

- 1.2 Summarize fundamental security concepts.

Materials Required

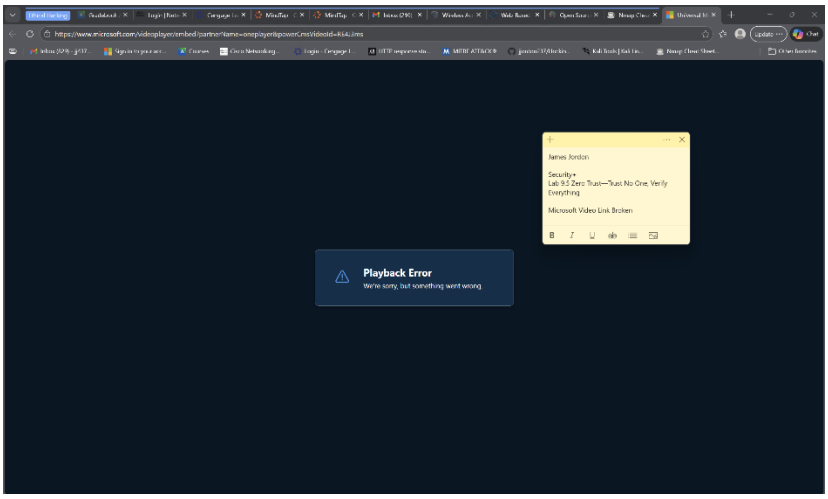
This lab requires the following:

- A computer running the Windows operating system with Internet access

Activity

In this lab, you search the Internet for information related to the Zero Trust model.

1. Open a browser and navigate to the following site and then watch the video: **https://www.microsoft.com/videoplayer/embed/RE4J3ms**



2. Navigate to the following site and then read the content: **https://www.cisa.gov/zero-trust-maturity-model**

3. Navigate to the following site and then read the content: **https://www.nist.gov/blogs/taking-measure/zero-trust-cybersecurity-never-trust-always-verify**.

4. After reading and reviewing the pages, you can download a copy of the version of the Applying Zero Trust Principles to Enterprise Mobility. Go to **https://www.cisa.gov/zero-trust-maturity-model** and scroll down to find the link to the PDF file.

5. Search the Internet for **Zero Trust Solutions** and navigate to the sites of some companies offering solutions.

Evaluate your current environment (school, work, home, or made-up environment) and write at least a one-page Zero Trust Solution document detailing how to implement this methodology. Also, add one to three of the companies that offer "Zero Trust Solutions."

Implementing a Zero Trust Security Model in a Home Environment

Introduction

As technology becomes increasingly integrated into daily life, home networks now resemble small enterprise environments. The household under evaluation consists of six family members—two parents and four children—using multiple connected devices including smartphones, laptops, iPads, an Xbox, and Nintendo Switch consoles. All devices connect to a shared home Wi-Fi network. While traditional security measures such as passwords and device-based protections may be in place, the absence of structured access control and network segmentation presents security risks. Implementing a Zero Trust security model can significantly enhance the protection of devices, personal information, and network resources within this environment.

Current Environment Assessment

The existing home network operates under a traditional perimeter-based trust model, meaning that any device connected to the Wi-Fi network is implicitly trusted. This configuration creates vulnerability if one device becomes compromised through phishing, malicious downloads, insecure gaming interactions, or weak passwords. Because all devices share the same network, a compromised system could potentially access sensitive information on other devices. Additionally, shared credentials or insufficient authentication practices may increase exposure to unauthorized access. The lack of centralized monitoring or continuous device verification further amplifies these risks.

Zero Trust Framework for the Home

The Zero Trust model is based on the principle "never trust, always verify." Rather than assuming devices inside the network are safe, Zero Trust requires continuous authentication and validation of users and devices before granting access to resources.

Identity and Access Management

The first step in implementing Zero Trust in the home environment is strengthening identity controls. Each family member should maintain individual accounts for major services (e.g., Microsoft, Apple, Google, gaming platforms). Multi-factor authentication (MFA) should be enabled for email, financial accounts, and other critical services. Using a password manager to generate and store strong, unique passwords reduces the likelihood of credential compromise. These measures ensure that access is based on verified identity rather than network location.

Network Segmentation

Network segmentation is essential to limit lateral movement between devices. A modern router that supports multiple Service Set Identifiers (SSIDs) or Virtual Local Area Networks (VLANs) should be configured to create separate networks. For example, one network can be designated for trusted devices such as parents' laptops and smartphones, another for children's devices and gaming consoles, and a third for guest access. Firewall rules should restrict communication between these segments. This structure prevents compromised gaming or mobile devices from accessing more sensitive systems.

Continuous Monitoring and Device Security

Zero Trust requires ongoing validation of device health. All devices should have automatic operating system and firmware updates enabled. Endpoint protection software should be installed on laptops, and parental controls should be applied where appropriate. Monitoring router logs and enabling alerts for unfamiliar devices can provide early detection of suspicious activity. These practices ensure that devices remain compliant and secure over time.

Leveraging Industry Solutions

While many Zero Trust principles can be implemented independently, partnering with established cybersecurity providers can significantly strengthen the security posture of a home network.

Cisco offers solutions such as Duo for multi-factor authentication and Umbrella for DNS-layer protection. Umbrella blocks malicious domains before devices connect, protecting gaming consoles and mobile devices from phishing and malware threats. Microsoft provides Microsoft Defender and Entra ID (formerly Azure Active Directory), which enhance identity management and endpoint security, particularly for Windows-based systems. Google's Advanced Protection Program and BeyondCorp principles emphasize identity-based access controls and device verification, which are especially beneficial for households using Android devices or Google services.

These companies provide enterprise-grade technologies that can be adapted to home environments, enhancing identity verification, device trust, and proactive threat prevention.

Conclusion

Implementing a Zero Trust model in a home environment significantly reduces the risk of unauthorized access and device compromise. By strengthening identity verification, segmenting the network, applying least-privilege access principles, and continuously monitoring device health, the household can create a secure and resilient digital environment. Leveraging established cybersecurity vendors further enhances protection through advanced authentication, endpoint security, and proactive threat detection. As home networks continue to expand in complexity, adopting a Zero Trust approach is a practical and forward-looking security strategy.